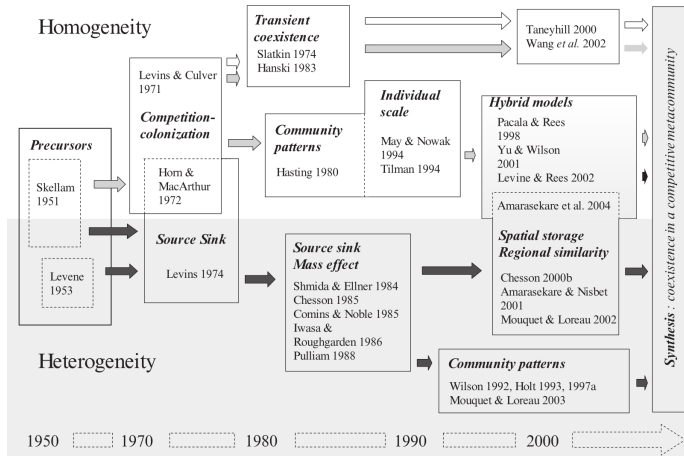




Université de Sherbrooke
Summer school in biodiversity modelling August 24th, 2026

A brief history of spatial ecology

Dominique Gravel

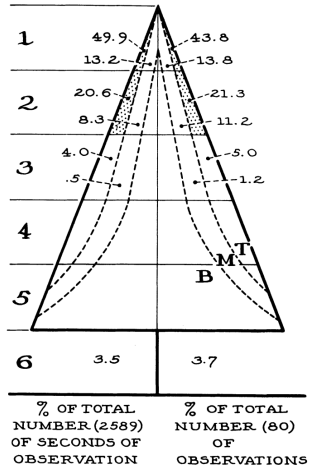


Mouquet et al. 2005 in Holyoak et al. Metacommunity Ecology. Chicago University Press

The world is patchy and heterogenous



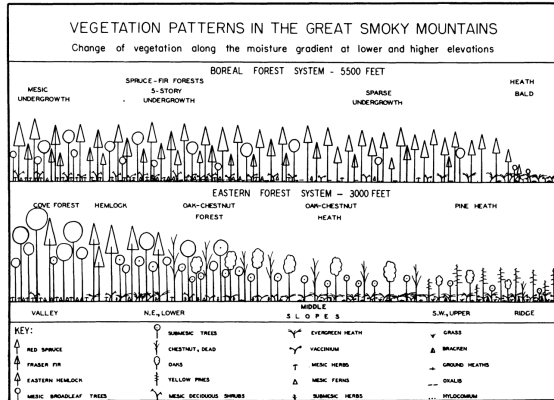
The niche concept



MacArthur, R.H. 1958. Ecology



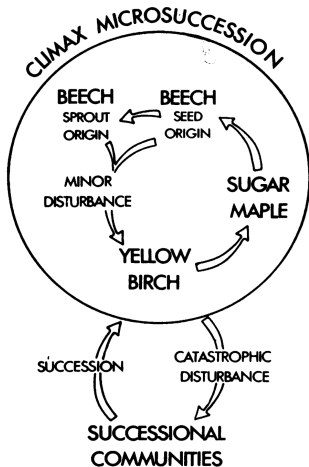
Species sorting



Whittaker, R.H. 1956. Ecol. Monogr.



Disturbances



Clements, 1936; Tansley, 1935; Watt, 1947; (Forcier, 1975)

Dispersal is limited



Diffusion theory

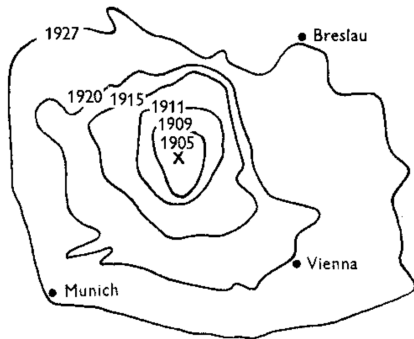


Fig. 1

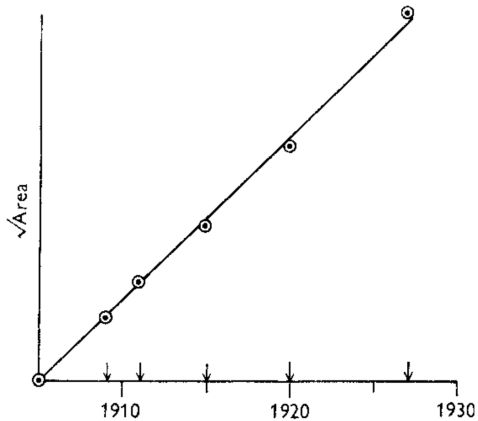
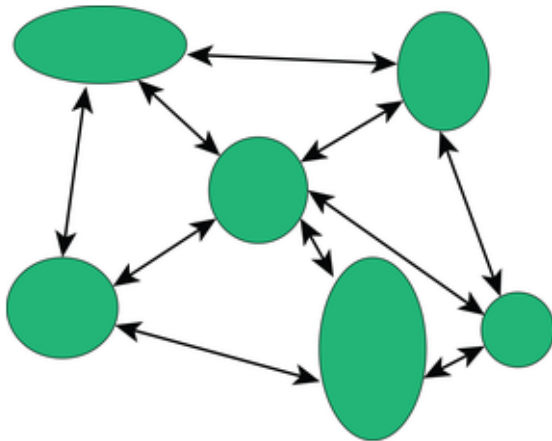


Fig. 2

Skellam, J.G. 1951. Biometrika.



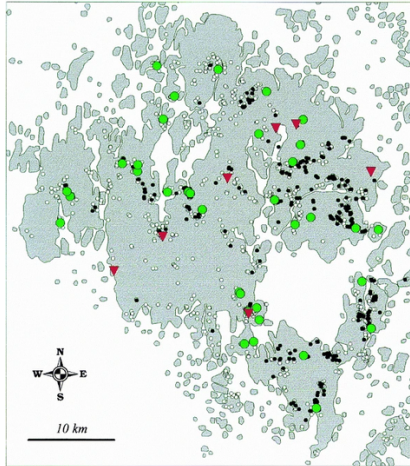
Patch occupancy



Levins, 1969. Bull. Entomo. Soc. Am.



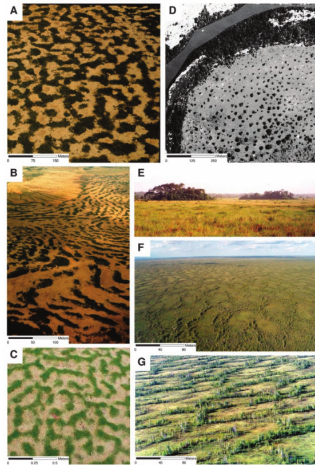
Spatially explicit landscapes



Hanski, I. 1994. J. Animal Ecol.



Patch formation



Solé et al. 1992. Phys. Lett. A (Rietkerk et al. 2004)

Local-regional coupling contributes to diversity



Island biogeography

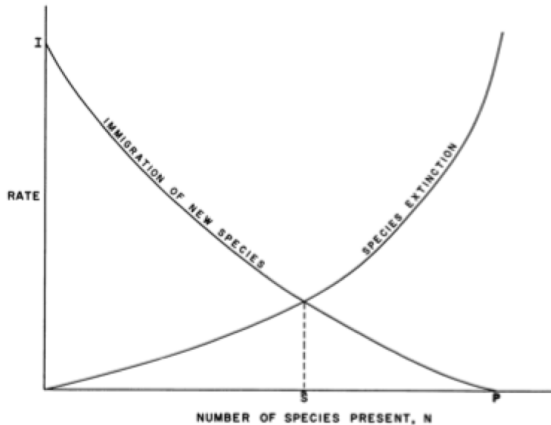
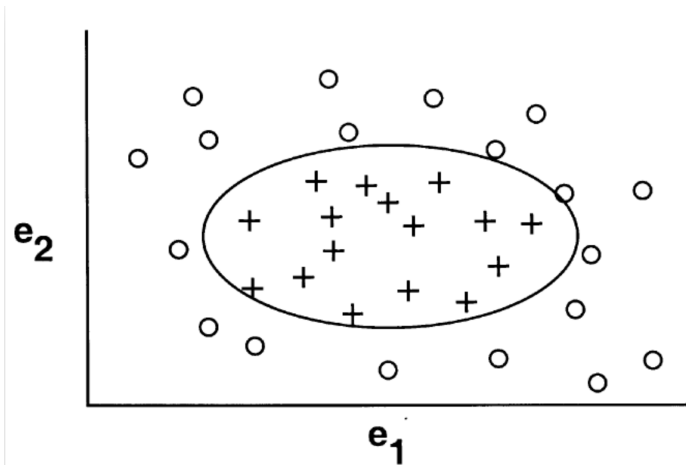


FIG. 4. Equilibrium model of a fauna of a single island. See explanation in the text.

MacArthur and Wilson. 1963. Evolution



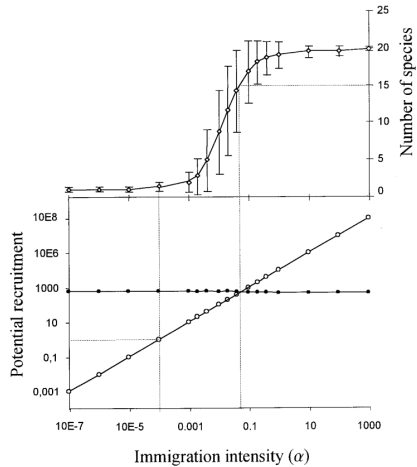
Source-sink-dynamics



Pulliam, R.H. 1988. Am. Nat.



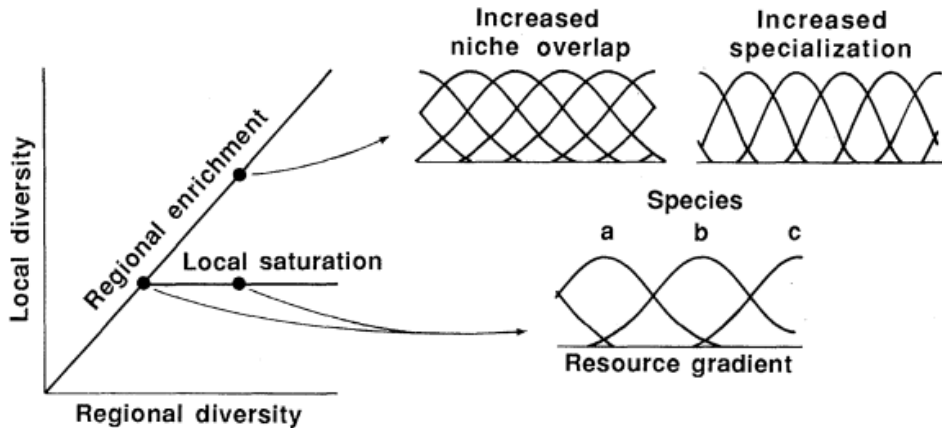
Mass effect



Loreau and Mouquet. 1999. Am. Nat.



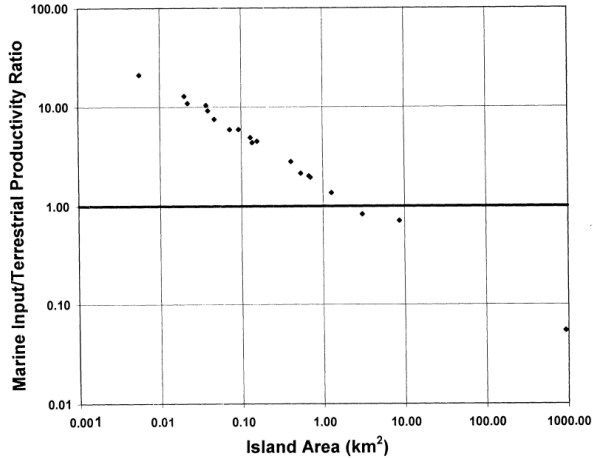
Local-regional diversity relationship



Ricklefs, R. 1987. Science.



Allochthonous inputs

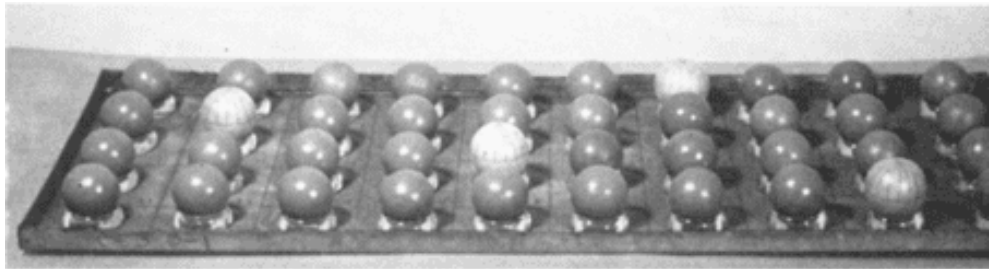


Polis, G. et al. 1997. Ann. Rev. Ecol. Sys.

Heterogeneity, dispersal & selection interact to impact community assembly



Arlequin dynamics and persistence



Huffaker, C.B. 1958. *Hilgardia*.



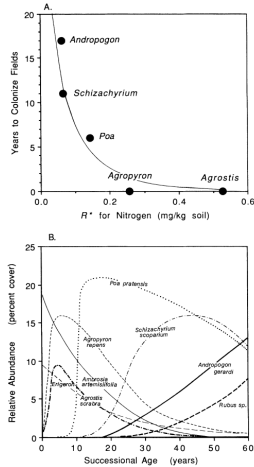
Priority effects



Hanski, I. 1983. Ecology.



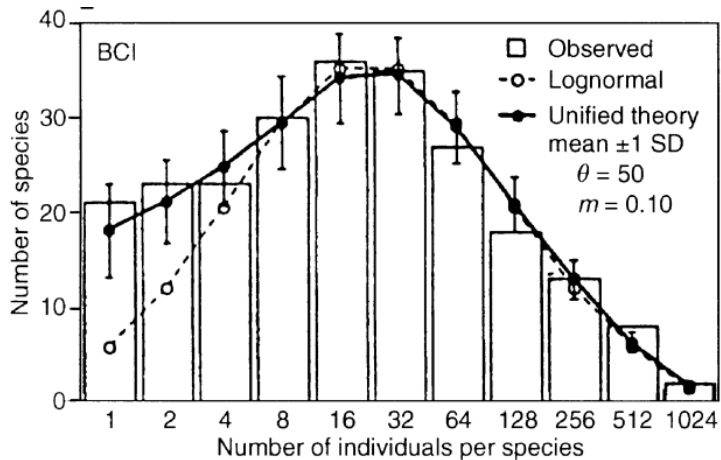
Competition-colonization trade-off



Hastings, 1980. Tilman, 1994.



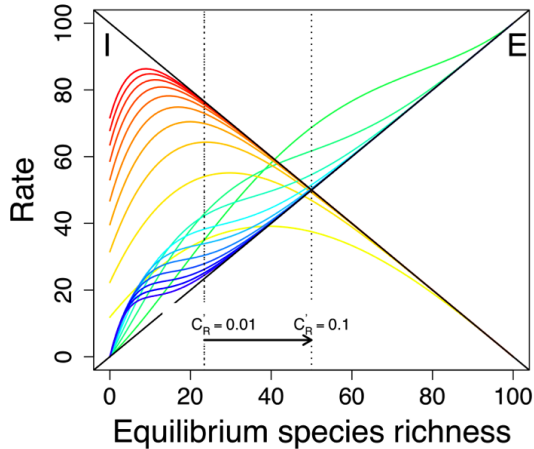
Neutral theory



Hubbell, 1997, 2001; Bell, 2000



Spatial inefficiency of predator-prey dynamics



Holt et al. 1999. Ecology ; Gravel et al. 2011. Eco. Lett.



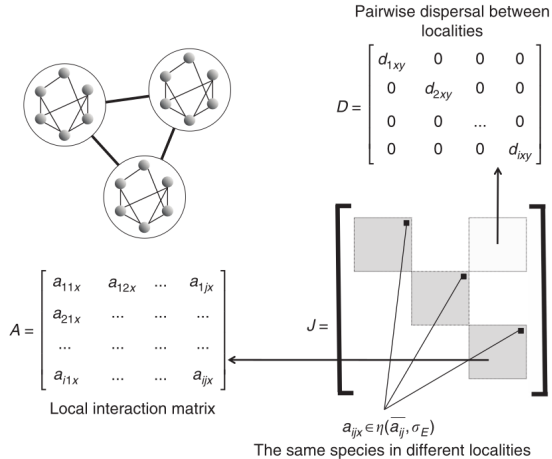
Spatial cascades



Loreau et al. 2003, *Eco. Lett.*



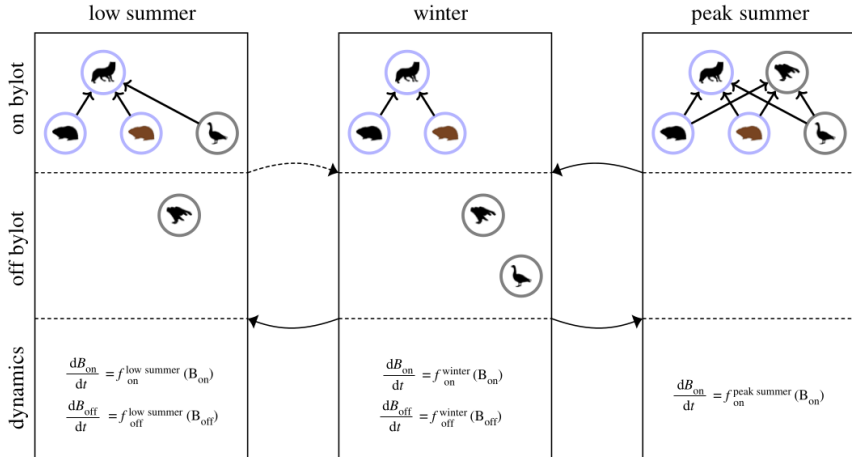
Stability-complexity



Gravel et al. 2016. Nat. Comm.



Distinct role of migration



Hutchison et al. 2020. Proc. Roy. Soc. London A

*Spatial ecology in the context
of global changes ?*